

EcoTrack: Smart Waste Management

CONFIDENTIAL

Service Blueprint & Operational Workflow Architecture

EXECUTIVE SUMMARY:

This service blueprint details the end-to-end interactions, touchpoints, backstage workflows, and automated system systems supporting the EcoTrack Smart Waste Management platform at the GreenTech Institute of Technology. The framework connects three primary roles (Admin, Collector, and Citizen) with physical IoT sensor telemetry to maintain campus hygiene and reduce operational overhead.

OPERATIONAL SERVICE BLUEPRINT MATRIX

SWIMLANES	1. IOT TELEMETRY	2. THRESHOLD ALERT	3. OPERATION COLLECT	4. CITIZEN ENGAGE
Physical Evidence	Smart Dustbins, Battery/Temp indicators, live dashboards.	LED warning lights on bins, Admin alerts tab, neon notifications.	Collector navigation map, QR Code, trash bags, collection trucks.	Public bin map list, report form screen, Eco-Point scorecard.
User Actions	Admin checks metrics. Citizen disposes waste.	Admin reviews alerts drawer, triage priority levels.	Collector follows route map, scans QR code to clear bin.	Citizen reports overflow via app, checks cleanest bins.
Onstage / Visible Contact Actions	None (automated background operations).	Push warning message to Collector terminals.	Collector manually empties bin at campus site.	Citizen inputs location & photos of overflow.
Backstage / Support Actions	IoT updates stored locally. Timers run.	Algorithms identify optimal pickup routes.	Database clears bin logs, resets status to normal.	Eco-points credited to citizen profile.
Support Processes	Periodic stream simulations update Riverpod state.	State notifier tracks alerts & updates metrics.	GoRouter handles details drill down page navigation.	Auth validations checks citizens permissions.

KEY SERVICE STAGES & ARCHITECTURE:

1. Real-Time Telemetry Node System:

The 25 campus nodes periodically transmit fill levels, battery charges, gas indicators, and temperatures. This keeps dashboards responsive, reducing physical inspections by 40%.

2. Smart Automated Routing & Triage:

When capacity hits >= 85%, the system elevates node status to "Overflowing". The collector dashboard reactively updates tasks, ranking bins based on proximity and breach urgency.

3. QR-Based Pick Validation:

Waste collection is validated by physical presence. The collector must scan the QR code fixed on the bin. Successful scan resets sensor registers and updates campus compliance metrics.

4. Gamified Citizen Reporting:

Citizens can report anomalies (e.g. overflow, bad odor) instantly. Dispatches trigger a prompt notification for manual collection while rewarding the reporting citizen with Eco-Points.

GreenTech Operations Team